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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc DEGREE EXAMINATION-APRIL 2021
BC507-MOLECULAR BIOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION A (25X1=25)
Answer ALL questions:

Choose the correct answer: (10X1=10)

1. DNA helix under normal conditions exist in

   a. A form b. B form c. Z form d. E form

2. The ratio of purine to pyrimidines in DNA is

   a. 0 b. 1 c. 2 d. 3

3. The protein that stabilizes ssDNA from renaturation is

   a. Helicase b. Gyrase c. SSB d. DNa A

4. The RNA primer in lagging strand is removed by

   a. RNA primase b. RNase c. DNA pol I d. DNa pol III

5. The cap structure in eukaryotic mRNA is

   a. 6-methylguanylate b. 9-dimethylguanylate

   c. 9- ethylguanylate d. 7- methylguanylate

6. The subunit of RNA polymerase used for promoter recognition is

   a. α b. β c. β' d. σ

7. Which is called as ochre codon

   a. UAA b. UAG c. UGA d. UGG

8. Attenuator sequence is present in

   a. Lac operon b. Trp operon c. Ara operon d. All of the above

9. Pyrimidine dimers are repaired by

   a. Base excision b. Photoreactivation

   c. Excision repair d. Mismatch repair

10. AP sites are formed in

    a. Direct repair b.Base excision c.SOS repair d.Mismatch repair

II. True or False (5X1=5)

11. Heterochromatin is transcriptionally active.
12. Intron codes for proteins.
13. tRNA has anticodon loop.
14. Genetic code is a triplet.
15. Rec A proteins are involved in recombination repair.

III Answer in one sentence (10X1=10)

16. Histones.
17. Heterochromatin.
18. Kornberg enzyme
19. Okasaki fragment
20. Polycistronic mRNA
21. SNurps
22. Stop codons
23. Operator site
24. Leading strand.
25. Pyrimidine dimer

SECTION-B (4X5=20)
Answer any FOUR of the following:

26. Explain the central dogma of molecular biology.
27. List out the factors involved in prokaryotic replication.
28. What are promoters? Bring out their importance.
29. Comment on rho dependent termination of prokaryotic transcription.
30. Discuss the role of DNA photolyase.
31. What is operon? Highlight the organization of lac operon genes.

SECTION–C (3X10=30)
Answer any THREE of the following:

32. Explain the organization of chromosomes.
33. How will you prove that DNA replication is semi-conservative?
34. Describe the events in post transcriptional modification of mRNA.
35. Explain protein synthesis in prokaryotes.
36. With a neat diagram, explain any two DNA repair process.

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